Venkata Suresh Kumar S, Ph.D.

Principal Bioinformatics Scientist

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PROFESSIONAL SUMMARY

Bioinformatics scientist and subject matter expert with 15+ years of experience in data analysis, product development, service delivery, NGS workflow automation, Containerization and working in onshore-offshore setting. My work has primarily focused on genomics (clinical and translational), transcriptomics (bulk, single-cell, and spatial), metagenomics, metabolomics, and the full SDLC of bioinformatics products. More recently, I've been interested in AI-augmented omics workflows, particularly automation, workflow execution, and multimodal integration.

CORE COMPETENCIES

Bioinformatics Product Development	Bioinformatics service delivery
BI and CG Agile SDLC	Bulk, Single cell and Spatial transcriptomics
Clinical Genomics	NGS workflows including AI augmented workflows
Workflow virtualization (Docker)	Project management (Atlassian tools)

PROFESSIONAL EXPERIENCE

Bioinformatics Scientist Apr 2022 - May 2026
AdvaitaBio Corporation • Ann Arbor, MI, USA (Remote)

- Served as the primary bioinformatics SME for iPathwayGuide™, iSCanGuide™, and iVariantGuide™ products for AdvaitaBio
- Involved in product design, statistical method implementation, testing and prototyping
- Integrated AI-assisted tools (Claude Code, Claude CoWork, Claude Skills) into bioinformatics workflow prototyping and automated API testing in Shell and Claude skill
- Handled multiple client projects for transcriptomics, metabolomics, and proteomics

Chief Bioinformatics Officer, Nov 2019 - Apr, 22
NavipointHealth India Private Limited • Hyderabad

- Defined and owned the bioinformatics vision and technical strategy for a health-tech startup, functioning as the sole senior scientific authority across all omics domains.
- Architected the DesignStudio™ platform, embedding scientific rigor and domain expertise directly into the product design.
- Established clinical genomics and gut microbiome service standards and analytical frameworks adopted across all client engagements.
- Led Bioinformatics Team for service and product delivery

Bioinformatics Manager Sep 2018 – Nov, 19
Bioserve Biotechnologies India Pvt. Ltd. (A Reprocell™ Company) • Hyderabad

- Established core services for Bioinformatics service projects (Hardware and Software)
- Handled Bioinformatics team for project deliveries
- Handled Prokaryotic, Eukaryotic, and Plant Bioinformatics projects including WGS, WES, miRNA, 16s, and mRNA data analyses.
- Led Bioinformatics Team for service delivery

Team Manager - Bioinformatics Jun 2016 – Nov, 16
Cognizant Technology Solutions India Pvt. Ltd Mumbai

- Applied bioinformatics expertise to biomarker discovery and analysis in neurological diseases

- Conducted functional network analysis to identify molecular pathways underlying neurological conditions.

Scientist II (Functional Lead) Jun 2013 – Sep, 14
Genome Lifesciences Private Limited (Parent: Genome International Corporation, WI, US) • Chennai

- Served as functional bioinformatics lead for Genome Explorer™ and IGR™ product development, ensuring scientific soundness of analytical features.
- Acted as Functional lead for Product development
- Bioinformatics data analysis services for clients in Europe and USA

Bioinformatics Analyst II Jul 2011 – May, 13,
SemanticBits India Pvt. Ltd. (Parent: SemanticBits, Herndon, US) • Hyderabad, India

- Applied bioinformatics domain expertise to US federal health informatics programmes, bridging scientific knowledge and software engineering requirements.
- Scientific contributions to the Clinical Genomicist Workstation led to co-authorship of the AMIA 2013 publication.

Senior Scientist II – Bioinformatics, Aug 2008 – Jun 2010,
Strand Lifesciences Private Limited • Bangalore

- Contributed bioinformatics domain knowledge to the development of GeneSpring GX™ and Mass Profiler Pro™, two industry-standard commercial platforms used globally by researchers.
- Provided scientific oversight for transcriptomics and statistical analysis engagements with pharma clients.

Post-Doctoral Research Fellow, 2005 – 2008,
Lerner Research Institute, Cleveland Clinic Foundation • Cleveland, OH, USA

- Developed deep expertise in cardiovascular transcriptomics and miRNA biology; findings contributed to a high-impact PLOS ONE publication on signaling networks in end-stage heart failure.
- Applied integrative bioinformatics approaches to mouse transgenic heart failure models, combining microarray analysis with pathway and network biology.
- Conducted sequence analysis and phylogenetic studies of GPCRs, building foundational expertise in structural bioinformatics.

PRODUCTS & PLATFORMS DELIVERED

- iPathwayGuide™, iSCanGuide™, iVariantGuide™ - AdvaitaBio, MI, USA
- DesignStudio™ - NavipointHealth Inc., India
- Genome Explorer™ & IGR™ - Genome Lifesciences / GIC, Chennai
- Clinical Genomicist Workstation™ - SemanticBits, Hyderabad (currently owned by Velsera)
- GeneSpring GX™, Mass Profiler Pro™ - Strand Lifesciences, Bangalore
- R-Shiny Apps: COVID-19 global/India dashboards, volcano plot tool, DNA/peptide sequence simulator

PUBLICATIONS

1. Parvez MK, Jagirdar RM, Purty RS, Venkata SK, *et al*. COVID-19 pandemic: Understanding the emergence, pathogenesis and containment (Review). World Acad Sci J 2:18, 2020.
2. Naga Prasad SV, Gupta MK, Duan ZH, Surampudi VSK, Liu CG, *et al*. A unique microRNA profile in end-stage heart failure indicates alterations in specific cardiovascular signaling networks. PLOS ONE 12(3): e0170456, 2017.
3. Surampudi S *et al*. Clinical genomicist workstation. AMIA Summits Transl Sci Proc. 2013; 2013:156-7.
4. S V Suresh Kumar *et al*. Combined sequence and structure analysis of the fungal laccase family. Biotechnology and Bioengineering, 83(4):386-394, 2003.
5. Punekar NS, Suresh Kumar SV, Jayashri TN, Anuradha R. Isolation of genomic DNA from acetone-dried Aspergillus mycelia. Fungal Genet Newsl. 2003; 50:15-16.

RESEARCH PROJECTS

1. miRNA signalling networks in end-stage human heart failure - Cleveland Clinic Foundation
2. Transcriptome profiling in transgenic heart failure mouse models - Cleveland Clinic Foundation
3. Sequence analysis & phylogenetics of G protein-coupled receptors (GPCRs)
4. Laccase purification, identification & characterization from Aspergillus niger - IIT Bombay
5. Sequence, structure & phylogenetic analysis of fungal and plant laccases - IIT Bombay
6. Molecular marker identification in *Colletotrichum graminicola* via RAPD/RFLP/AFLP - ICRISAT

EDUCATION

Ph.D., Biotechnology	IIT Bombay, Mumbai, India1999 - 2005
M.Sc., Biotechnology	University of Calicut, Calicut, India1996 - 1998
B.Sc., BZC	Sri Venkateswara University, Tirupati, India1993 - 1996

FELLOWSHIPS & ACADEMIC DISTINCTION

1. DBT (Dept. of Biotechnology, India) Fellowship in M.Sc. Biotechnology, 1996–98
2. CSIR-JRF & GATE-98 qualified for Life Sciences
3. University 3rd Rank, M.Sc. Biotechnology - University of Calicut
4. District 1st in Science Fair & Science Quiz, Andhra Pradesh (1989)

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